

Brahadeesh Sankarnarayanan

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RESEARCH INTERESTS

I work in extremal combinatorics and graph theory. My current interests are in spectral extremal graph theory, intersection graphs of paths, intersecting set systems, graph convexity, and majority coloring games. I have also worked on colorings and list colorings of graphs embedded in surfaces, and on the ranks of structured matrices arising from tournaments.

Current Projects

Majority coloring game on trees

With Yash Chawda (IIT Jodhpur), Saraswati Girish Nanoti (IISc Bangalore) and Eshwar Srinivasan (Amrita Vishwa Vidyapeetham, Coimbatore). Investigating optimal strategies and computational complexity for the two-player majority coloring game, particularly on binary and ternary trees.

$\triangle$ -convexity in graphs

With Vishnu Kumar (IIT Jodhpur). Investigating convexity structures in graphs, particularly computing dimensional parameters such as the exchange number and Carathéodory number for $\triangle$ -convexity, and characterizing the extremal examples.

Brualdi–Hoffman-type extremal problems

With M. Rajesh Kannan (IIT Hyderabad). Investigating spectral extremal problems for F -free graphs on m edges, extending recent results for the case $F = K_3$ to $F = K_4$ and beyond.

Intersection graphs of paths

With Niranjan Balachandran (IIT Bombay), Shagnik Das (National Taiwan University) and Umesh Shankar (IISc Bangalore). Investigating the path-representation number $\text{prn}(G)$ of a graph G : bounds in terms of other graph parameters, algorithms for computing optimal path representations, computational complexity of the parameter for different graph classes.

Fractional intersecting families

Continuing investigation of the Balachandran–Mathew–Mishra conjecture on the maximum size of fractional θ -intersecting families over $[n]$, building on [Balachandran–Das–Sankarnarayanan \(2025\)](#) and [Balachandran–Bhattacharya–Kher–Mathew–Sankarnarayanan \(2023\)](#).

EDUCATION

PhD (Mathematics)

2018 – 2024

Indian Institute of Technology (IIT) Bombay

Thesis: “Some problems in combinatorics: Excursions in graph colorings and extremal set theory”

Advisor: Niranjan Balachandran

CPI: 8.89/10.0

BS–MS (Dual) Degree (Major in Mathematics)

2012 – 2017

Indian Institute of Science Education and Research (IISER) Bhopal

Thesis: “Automorphic forms and Tate’s thesis”

Advisor: Karam Deo Shankhadhar

CPI: 9.31/10.0

EMPLOYMENT

Assistant Professor (Grade II)

09/2025 – present

Department of Mathematics, IIT Jodhpur

Research Associate

05/2025 – 09/2025

Department of Mathematics, IIT Hyderabad

Project: “Spectral Theory of Signed Graphs”

Principal Investigator: M. Rajesh Kannan

Institute Post-Doctoral Fellow

Department of Mathematics, IIT Bombay

Mentor: Niranjan Balachandran

08/2024 – 02/2025

Project Assistant

Department of Mathematics, IISER Bhopal

Project: “Self-dual representations and signs”

Principal Investigator: Kumar Balasubramanian

09/2017 – 11/2017

PUBLICATIONS

Refereed Journal Articles

- [J9] *5-list coloring toroidal 6-regular triangulations in linear time* (with N. Balachandran). Discrete Applied Mathematics **386** (2026), 75–103, [doi:10.1016/j.dam.2026.01.032](https://doi.org/10.1016/j.dam.2026.01.032), [arXiv:2106.01634](https://arxiv.org/abs/2106.01634), [MR5028701](https://mrindex.org/entry/MR5028701), [Zbl 08187875](https://zbmath.org/journals/DAM/Balachandran/202601032).
- [J8] *Bounded fractional intersecting families are linear in size* (with N. Balachandran and S. Das). The Electronic Journal of Combinatorics **32** (2025), no. 3, #P3.34, 10 pp., [doi:10.37236/12900](https://doi.org/10.37236/12900), [arXiv:2402.14981](https://arxiv.org/abs/2402.14981), [MR4949605](https://mrindex.org/entry/MR4949605), [Zbl 1572.05240](https://zbmath.org/journals/EJC/Balachandran/202503034).
- [J7] *Low-rank matrices, tournaments, and symmetric designs* (with N. Balachandran). Linear Algebra and its Applications **694** (2024), 136–147, [doi:10.1016/j.laa.2024.04.006](https://doi.org/10.1016/j.laa.2024.04.006), [arXiv:2401.14015](https://arxiv.org/abs/2401.14015), [MR4736676](https://mrindex.org/entry/MR4736676), [Zbl 1539.05071](https://zbmath.org/journals/LAA/Balachandran/202404006).
- [J6] *Almost full rank matrices arising from transitive tournaments* (with N. Balachandran and S. Bhattacharya). Linear and Multilinear Algebra **72** (2024), no. 2, 153–161, [doi:10.1080/03081087.2022.2158168](https://doi.org/10.1080/03081087.2022.2158168), [MR4703565](https://mrindex.org/entry/MR4703565), [Zbl 1533.05147](https://zbmath.org/journals/LMA/Balachandran/202402002).
- [J5] *On hierarchically closed fractional intersecting families* (with N. Balachandran, S. Bhattacharya, K. V. Kher and R. Mathew). The Electronic Journal of Combinatorics **30** (2023), no. 4, #P4.37, 17 pp., [doi:10.37236/11651](https://doi.org/10.37236/11651), [arXiv:2211.02540](https://arxiv.org/abs/2211.02540), [MR4672585](https://mrindex.org/entry/MR4672585), [Zbl 1533.05268](https://zbmath.org/journals/EJC/Balachandran/202304037).
- [J4] *An ensemble of high-rank matrices arising from tournaments* (with N. Balachandran and S. Bhattacharya). Linear Algebra and its Applications **658** (2023), 310–318, [doi:10.1016/j.laa.2022.11.004](https://doi.org/10.1016/j.laa.2022.11.004), [arXiv:2108.10871](https://arxiv.org/abs/2108.10871), [MR4513895](https://mrindex.org/entry/MR4513895), [Zbl 1506.15034](https://zbmath.org/journals/LAA/Balachandran/202211004).
- [J3] *Note on 4-coloring 6-regular triangulations on the torus*. Annals of Combinatorics **26** (2022), no. 3, 559–569, [doi:10.1007/s00026-022-00573-8](https://doi.org/10.1007/s00026-022-00573-8), [arXiv:2106.01037](https://arxiv.org/abs/2106.01037), [MR4473892](https://mrindex.org/entry/MR4473892), [Zbl 1497.05098](https://zbmath.org/journals/ANCO/Balachandran/202203003).
- [J2] *The choice number versus the chromatic number for graphs embeddable on orientable surfaces* (with N. Balachandran). The Electronic Journal of Combinatorics **28** (2021), no. 4, #P4.50, 20 pp., [doi:10.37236/10263](https://doi.org/10.37236/10263), [arXiv:2102.06993](https://arxiv.org/abs/2102.06993), [MR4394670](https://mrindex.org/entry/MR4394670), [Zbl 1486.05101](https://zbmath.org/journals/EJC/Balachandran/202104050).
- [J1] *Self-dual representations of $SL(2, F)$: an approach using the Iwahori–Hecke algebra* (with K. Balasubramanian and K. S. Senthil Raani). Communications in Algebra **47** (2019), no. 10, 4210–4215, [doi:10.1080/00927872.2019.1581211](https://doi.org/10.1080/00927872.2019.1581211), [MR3975999](https://mrindex.org/entry/MR3975999), [Zbl 1432.20031](https://zbmath.org/journals/COMA/Balasubramanian/201910010).

Conference Proceedings

- [C2] *On the majority game chromatic number of forests and other graphs* (with Y. Chawda, S. G. Nanoti and E. Srinivasan). Accepted at FSTTCS 2026. [arXiv:2609.23803](https://arxiv.org/abs/2609.23803).
- [C1] *5-list coloring toroidal 6-regular triangulations in linear time* (with N. Balachandran). Extended abstract. CALDAM 2023. In: A. Bagchi and R. Muthu (eds.) *Algorithms and Discrete Applied Mathematics*, Lecture Notes in Computer Science, vol. 13947, pp. 134–146, Springer, Cham (2023), [doi:10.1007/978-3-031-25211-2_10](https://doi.org/10.1007/978-3-031-25211-2_10), [MR4573050](https://mrindex.org/entry/MR4573050), [Zbl 07728658](https://zbmath.org/journals/LNCS/Balachandran/202305010).
The full version appeared as [J9] above.

Manuscripts in Preparation

[P1] *An extremal problem for vertex intersection graphs of paths* (with N. Balachandran, S. Das and U. Shankar). 2025, in preparation.

RESEARCH SUPERVISION

PhD Students

Piyush Joshi Department of Mathematics, IIT Jodhpur	enrolled Autumn 2026–27
Yash Chawda Department of Mathematics, IIT Jodhpur	enrolled Spring 2025–26
Vishnu Kumar Department of Mathematics, IIT Jodhpur	enrolled Autumn 2025–26

Executive MTech (Data and Computational Sciences) Projects

Rayipalli Prakash School of Artificial Intelligence and Data Science, IIT Jodhpur Project: “GNN-Based Automatic Calorie Identification from Food Images”	2025–26
Md Parvez Rahaman School of Artificial Intelligence and Data Science, IIT Jodhpur Project: “AI-Based Text Summarization and Tone Rewriting System Using BART and T5”	2025–26

Summer Interns: SURAJ Program

Vaibhav Suvagiya Sardar Vallabhbhai National Institute of Technology, Surat Project: “RAMO: A Verified MaxSAT Repair Oracle for Tabu Search on Ramsey 2-Colouring, A De-Confounded Negative Result”	Summer 2026
B. James Mejulan Sardar Vallabhbhai National Institute of Technology, Surat Project: “Generators and Relations for Discrete Groups”	Summer 2026

Other Interns

Sai Akshar Tadimeti IIT Jodhpur Reading Project: Discrete mathematics, from Rosen’s <i>Discrete Mathematics and Its Applications</i>	Summer 2026
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FELLOWSHIPS, AWARDS & HONORS

Institute Post-Doctoral Fellowship IIT Bombay	08/2024 – 02/2025
National Board for Higher Mathematics (NBHM) Research Award Department of Atomic Energy, Government of India	2018 – 2023
Joint CSIR–UGC National Eligibility Test for Junior Research Fellowship CSIR – Human Resource Development Group; All-India Rank 20 in Mathematical Sciences	2017
Director’s Gold Medal IISER Bhopal Awarded for outstanding all-round achievement and leadership in the BS–MS graduating class	2017
Rashtrapati Bhavan In-Residence Program for Scholars from IITs, IISERs, and IISc Invited by the President of India to represent IISER Bhopal	2016
Science Academies’ Summer Research Fellowship Indian Academy of Sciences, Indian National Science Academy, and National Academy of Sciences, India	2015 & 2016
World Science Conference, Israel	2015

Hebrew University of Jerusalem; member of a 15-person Indian contingent

INSPIRE Scholarship for Higher Education

2012 – 2017

Department of Science & Technology, Government of India

RESEARCH GRANTS

Research Initiation Grant (RIG)

2026 – 2027

IIT Jodhpur

Amount: Rs. 4,00,000

Role: Principal Investigator

INVITED TALKS AND PARTICIPATION

Technical Talks

Teaching logical statements at the undergraduate level: Why one common definition gets in the way 2026

Conference on the Practice of Teaching Undergraduate Mathematics, Krea University, Sri City

Intersection Graphs and Path Representations 2026

15th Prof. E. Sampathkumar Memorial Endowment Lecture, 22nd Annual Conference of the ADMA & International Conference on Discrete Mathematics, IIT Madras

Tight bounds on the Carathéodory and exchange numbers for Δ -convexity spaces 2026

Annual Meru Combinatorics Conference, IIT Bhilai

Competency-based Learning in Mathematics 2026

Session at the STEM Teacher Training Program, organized (in association with Jodhpur Zila Parishad and Shiksha Vibhag) by the Centre for Education Technology, IIT Jodhpur

Nonstandard analysis: a gentle introduction 2025

Discussion Meeting on Number Theory and Combinatorics, Evolve Back, Coorg

Bisection-closed families, tournaments, and symmetric designs 2025

Geometry, Probability, and Algorithms, International Centre for Theoretical Sciences, Bengaluru

Sunflowers and symmetric designs 2025

Research Seminar, Department of Mathematics, IIT Jodhpur

Sunflowers and symmetric designs 2024

Research Associate Seminar, Theoretical Statistics and Mathematics Unit, Indian Statistical Institute, Bangalore Centre

Some recent work on fractional intersecting families 2024

Annual Meru Combinatorics Conference, Graphic Era Hill University, Bhimtal Campus

Sunflowers, tournaments, and symmetric designs 2024

Institute Post-Doctoral Fellowship Seminar, Department of Mathematics, IIT Bombay

Some recent work on fractional intersecting families 2024

CSE Research Symposium, IIT Palakkad

Popular and Expository Talks

What Is Combinatorial Thinking? 2026

Summer School for Women in Mathematics and Statistics, ICTS Bengaluru

A Combinatorial Gem of Maryam Mirzakhani: List Colorings of Planar Graphs 2026

One Day Symposium Celebrating Women in Mathematics, IIT Jodhpur

Counting the Infinite 2025

National Mathematics Day Guest Lecture, Government College Makroniya Bujurg, Sagar, Madhya Pradesh (online)

A Glimpse of Infinity 2020

National Mathematics Day Webinar, Raja Bhoj Govt. College Katangi, Balaghat, Madhya Pradesh (online)

Research Discussion Groups

Leader, Majority coloring game on trees	2026
MATRIX-MFO Tandem Workshop: Combinatorial Interchange, MATRIX, Creswick, Australia	
Participant, List packing conjecture	2026
MATRIX-MFO Tandem Workshop: Combinatorial Interchange, MATRIX, Creswick, Australia	
Leader, Extremal problems on fractional intersecting families	2026
Conference on Frontiers in Combinatorics – Topology, Algebra and Data-Driven Insights, BITS Pilani, Pilani Campus	

EVENTS ORGANIZED

Screening of <i>Secrets of the Surface: The Mathematical Vision of Maryam Mirzakhani</i>	2026
Documentary film by George Csicsery; screening organized under the Mathematics Society as a follow-up to the One Day Symposium Celebrating Women in Mathematics Department of Mathematics, IIT Jodhpur	
Career Opportunities Beyond Academia	2026
One-day interaction session with industry mathematicians Department of Mathematics, IIT Jodhpur	
An Introductory Workshop on the Lean Theorem Prover	2026
Three-day workshop led by T. V. H. Prathamesh (Krea University); co-organized with D. Kaur and K. Goel Department of Mathematics, IIT Jodhpur	

TEACHING EXPERIENCE

Instructor: IIT Jodhpur

MAL7060 Statistics for Data Science	Autumn 2026–27
MAL6150 Probability and Statistics	Autumn 2026–27
MAL3040 Statistics	Autumn 2026–27
MAL2040 Design and Analysis of Experiments	Spring 2025–26
49 students; overall instructor rating 3.70/5.0 (mean), median 4 across all dimensions	

Teaching Assistant: IIT Hyderabad

MA5020 Functional Analysis	Autumn 2025–26
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Teaching Assistant: IIT Bombay

SI 419 Combinatorics	Autumn 2024–25, Autumn 2021–22
SI 416 Optimization	Spring 2021–22
MA 205 Complex Analysis	Autumn 2021–22, Autumn 2020–21, Autumn 2019–20
MA 207 Differential Equations II	Autumn 2020–21, Autumn 2019–20
MA 108 Differential Equations	Summer 2020–21, Spring 2019–20
MA 106 Linear Algebra	Spring 2020–21, Spring 2019–20
MA 111 Calculus II	Spring 2020–21

Other Teaching and Mentoring

Summer School for Women in Mathematics and Statistics (Resource Person)	2026
International Centre for Theoretical Sciences, Bengaluru	
Masterminds Program (Trainer)	2023–24, 2024–25
UAE IMO selection camp, Ministry of Education, UAE Topics: combinatorics and number theory	
Self-Assessment Team (Member)	2023
Mathematics Training and Talent Search Programme (MTTS)	

Online Foundation Course in Mathematics (Mentor) Mathematics Training and Talent Search Programme (MTTS)	2023
Online Short Course in Linear Algebra: Application View (Mentor) Mathematics Training and Talent Search Programme (MTTS)	2023
Study-Help Room (Co-organizer) Department of Mathematics, IIT Bombay Weekly drop-in math help for BS and MSc students	2023
TA Skill Enhancement & Training (Master TA) Parimal and Pramod Chaudhari Centre for Learning and Teaching, IIT Bombay	2022
A Basic Course in Number Theory (TA) NPTEL Online Certification course	2020

PROFESSIONAL SERVICE

Memberships

Life Member, Academy of Discrete Mathematics and Applications	2026 – present
Associate Fellow, The Institute of Combinatorics and its Applications	2026 – present
Life Member, The Indian Mathematical Society	2026 – present
Life Member, Mathematics Teachers' Association (India)	2026 – present
Affiliate Member, American Mathematical Society	2026 – present

Reviewing

Theoretical Computer Science	2026 – present
Indian Journal of Pure and Applied Mathematics	2025 – present
Acta Informatica	2025 – present
Proceedings – Mathematical Sciences (Indian Academy of Sciences)	2024 – present
Mathematical Reviews / MathSciNet	2023 – present
zbMATH Open	2022 – present
Conference on Algorithms and Discrete Applied Mathematics (CALDAM)	2020 – present

Session Chair

Keynote session	2026
K. Kant (ABV-IIITM Gwalior): <i>Projection methods for Volterra and Fredholm integral equations with mixed weakly singular kernels</i>	
A. Jha (IIT Gandhinagar): <i>Domain decomposition methods for implicit solvation models</i>	
Workshop on Advanced Methods in Integral Equations and Their Applications, IIT Jodhpur	
5th Prof. R. Balakrishnan Endowment Lecture	2026
R. Venkatesh (IISc): <i>How to obtain the Tutte polynomials of graphs from representation-theoretic methods?</i>	
22nd Annual Conference of the ADMA & International Conference on Discrete Mathematics, IIT Madras	

INSTITUTIONAL SERVICE

IIT Jodhpur

Member, Department Postgraduate Committee (DPGC) Department of Mathematics	2026 – present
Faculty Advisor, MTech (Data and Computational Sciences) Batch of 2026–2028, Department of Mathematics	2026 – present
Member, Department-level committee for projection of new Exec. BS program Executive BS in Mathematics and Computing, Department of Mathematics	2026 – present

Member, Department-level committee for projection of new BS program BS in Mathematics and Computing, Department of Mathematics	2025 – present
Faculty Advisor, Mathematics Society Board of Departmental Societies	2025 – 2026

IIT Bombay

Coordinator, Students' Seminar Department of Mathematics	01/2023 – 05/2023
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IISER Bhopal

President, Students Activity Council	2016 – 2017
Coordinator, Mathematics Club	2013 – 2015

CONFERENCES, WORKSHOPS & EXTERNAL COURSES ATTENDED

MATRIX–MFO Tandem Workshop: Combinatorial Interchange MATRIX, Creswick, Australia (held simultaneously with MFO, Oberwolfach)	2026
5th International Conference on Enumerative Combinatorics and Applications University of Haifa, Israel (online)	2026
Conference on Frontiers in Combinatorics – Topology, Algebra and Data-Driven Insights BITS Pilani, Pilani Campus	2026
Indian School on Logic and Applications (ISLA) Krea University, Sri City	2026
Annual Meru Combinatorics Conference IIT Bhilai	2026
LeanLang for Verified Autonomy Hackathon Emergence Research Lab, IISc Bangalore	2026
Indo-Polish Pre-Conference School on Algorithms and Combinatorics IIT Dharwad	2026
Discussion Meeting on Number Theory and Combinatorics Evolve Back, Coorg	2025
Topics in Geometric Combinatorics (NCM Workshop) Chennai Mathematical Institute	2025
Annual Meru Combinatorics Conference BITS Pilani, K. K. Birla Goa Campus	2025
Geometry, Probability, and Algorithms International Centre for Theoretical Sciences, Bengaluru	2025
HDXs and Codes International Centre for Theoretical Sciences, Bengaluru	2025
Lean for the Curious Mathematician International Centre for Theoretical Sciences, Bengaluru	2025
Sparse graphs: treewidth, planarity, bounded expansion (GIAN) IIT Dharwad	2025
Combinatorial Methods in Enumerative Algebra International Centre for Theoretical Sciences, Bengaluru	2024
Discrete Integrable Systems International Centre for Theoretical Sciences, Bengaluru	2024
Annual Meru Combinatorics Conference Graphic Era Hill University, Bhimtal Campus	2024
A First Second Course in the Probabilistic Method	2024

IIT Palakkad

Annual Meru Combinatorics Conference

2023

Pondicherry University

Conference on Algorithms and Discrete Applied Mathematics (CALDAM)

2023

Dhirubhai Ambani Institute of Information and Communication Technology, Gandhinagar

Indo-Dutch Pre-Conference School on Algorithms and Combinatorics

2023

Dhirubhai Ambani Institute of Information and Communication Technology, Gandhinagar